armfield

RE14 – Advanced Fuel Cell Technology

The RE14 Advanced Fuel Cell Technology system provides students with a self-contained modular system, covering current fuel cell technology on a laboratory scale.

Using the system students can undertake experiments covering working principles, efficiency, and characteristics curves of electrolysers and fuel cells.

In addition to a PEM-fuel cell, it also contains an ethanol-fuel cell to compare the different technologies.

PRODUCT CODE: **RE14**

Categories: Educational, RE Series - Renewable Energy, RE14 Fuel Cell Technology

Tags: Electrolysers , Ethanol-Fuel Cell , Fuel Cell Stacks , Fuel Cell Technology , Fuel Cells , H2 Charger , H2 Storage , Hydrogen , PEM-Fuel Cell , RE Series , RE14 , Renewable Energy

DESCRIPTION

The RE14 Advanced Fuel Cell Technology system provides students with a self-contained modular system, covering current fuel cell technology on a laboratory scale.

Using the system students can undertake experiments covering working principles, efficiency, and characteristics curves of electrolysers and fuel cells. In addition to a PEM-fuel cell, it also contains an ethanol-fuel cell to compare the different technologies.

Including three PEM-fuel cells the kit enables students to build fuel cell stacks. H_2 Charger and H_2 Storage are provided with the system to allow for the easy generation and storage of hydrogen.

TECHNICAL SPECIFICATIONS

- 1 x Base unit large
- 1 x Potentiometer module
- 1 x Motor module without gear
- 1 x Solar module 2.5V, 420mA
- 1 x H_2 Charger
- 1 x H_2 Storage
- 1 x Gas storage module
- 3 x PEM-fuel cell module
- 1 x Electrolyzer module
- 1 x Ethanol fuel cell module
- 1 x Propeller
- 1 x 0,15 x Silicone tube 2mm
- 1 x Lamp with table clamp
- 1 x Safety test lead, 50cm, red
- 1 x Safety test lead, 50cm, black
- 2 x Safety test lead, 25cm, red
- 2 x Safety test lead, 25cm, black
- 2 x Digital multimeter
- 1 x Layout diagram
- 1 x SOFC-fuel cell module Storage
- 1 x Gas storage module
- 3 x PEM-fuel cell module
- 1 x Electrolyzer module
- 1 x Ethanol fuel cell module
- 1 x Propeller
- 1 x 0,15 x Silicone tube 2mm
- 1 x Lamp with table clamp
- 1 x Safety test lead, 50cm, red
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- 1 x Layout diagram
- 1 x SOFC-fuel cell module

FEATURES & BENEFITS

- Comprehensive experimentation system on fuel cell technology
- Two different fuel cell technologies: PEM and ethanol fuel cells
- Buildable fuel cell stacks with three PEM fuel cells
- Easy hydrogen generation and storage with H_2 Charger and H_2 Storage
- Laboratory scale
- Modular design
- Supplied in a self-contained aluminium case
- Includes in-depth manual and predefined experiments

Experimental content

- Explore the structure of the electrolyser and different fuel cells
- Properties of the electrolyser
- Properties of a PEM fuel cell
- Properties of an ethanol fuel cell
- Faraday and energy efficiency of the electrolyser and PEM fuel cell
- Parallel and series connection of PEM fuel cells
- Properties of a SOFC fuel cell
- Hydrogen storage with the $\mathbf{H_2}$ storage technology

Related curriculums

- Chemical Engineering
- Environmental Energies
- Renewable Energies

Other products in the series

- Storage Temperature: -10°C to +70°C
- Operating temperature range: +10°C to +50°C
- Operating relative humidity range: 0 to 95%, non-condensing

Operational conditions

- Storage Temperature: -10°C to +70°C
- Operating temperature range: +10°C to +50°C
- Operating relative humidity range: 0 to 95%, non-condensing

Requirements

Electrical supply: 110-230V AC 50-60Hz

- Level and stable work surface

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

- **Volume:** 0.038m³
- **Gross weigh:** 10Kg

Dimensions

TRAY

- **Length:** 0.640m

- **Width:** 0.165m
- **Height:** 0.370m

Order Codes

RE14: Advanced Fuel Cell Technology

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